CM617: SUSTAINABILITY PRINCIPLES AND PRACTICES
IIT BOMBAY

UNWRAP:
Unpacking Whether Replacement Alternatives Perform

Does sustainable packaging actually reduce carbon?
A paired meta-analysis of packaging life cycle assessments under realistic collection rates

Course Project Report

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1. Introduction

Background. Replacing plastic with glass, aluminium or cartons is widely assumed to cut carbon, but life cycle assessments (LCAs) conflict: a review of 53 LCAs found plastics were often not the worst option and that end-of-life (EoL) assumptions shaped the result (Dolci et al. 2025).

Gap. Almost all comparisons assume European or US waste systems. India differs: about 81% of waste is collected and open burning is common (Cottom et al. 2024); the main Indian beverage LCA used collection-for-recycling rates of 85% (aluminium), 70% (PET), 60% (glass) and 54% (carton) (TERI 2022). No study has tested whether an alternative's carbon advantage survives such rates.

Question. Does the reported GWP (100-year global warming potential) advantage of sustainable packaging survive pooling across studies and Indian collection rates, and at what collection rate does it disappear? We (i) pool paired published comparisons and (ii) recompute the main beverage containers under European and Indian rates.

2. Methods

Stage 1: paired meta-analysis. We searched open sources for LCAs giving numeric GWP for a conventional and an alternative container on the same functional unit. 27 pairs from 13 studies (42 distinct GWP values) qualified; 23 pairs from 11 studies have a plastic baseline and are pooled. Values came from abstracts, summaries or stated percentage differences (derivation recorded per row); ten were second-hand from one literature table. Effect size is $\ln(\mathrm{GWP}_{alt}/\mathrm{GWP}_{conv})$; pairing inside a study cancels differences in boundary, database and author choices. Pairs from one study were averaged, then studies were pooled with equal weight (within-study variances are not reported), giving a random-effects-style mean, a t-interval, the between-study SD τ, and study-level bootstrap intervals (20,000 resamples; with $\leq$3 studies per group these are close to ranges). Studies were coded by alternative material and academic vs. industry-linked funding. EoL and allocation choices were not reported consistently enough to code.

Stage 2: recomputation under collection rates. A screening model gives kg CO_2e per litre for a 500 mL aluminium can, 600 mL PET bottle and 650 mL single-use glass bottle (TERI masses and distances): production + conversion + transport − (collection × yield × recycling credit) + open burning of uncollected PET. Emission factors come from IAI, LSE, FEVE and Hopewell et al. (sheet S3 of the workbook); three unsourced overheads (can making, bottle conversion, truck factor) and the burned share of uncollected PET (10–80%) were varied over stated ranges. Monte Carlo (N=20,000, triangular distributions) was run for European defaults (PET 48%, glass 80%, aluminium 71%) and Indian rates (TERI; PET also at 90%). The crossover c^* is the aluminium collection rate at which aluminium equals PET.

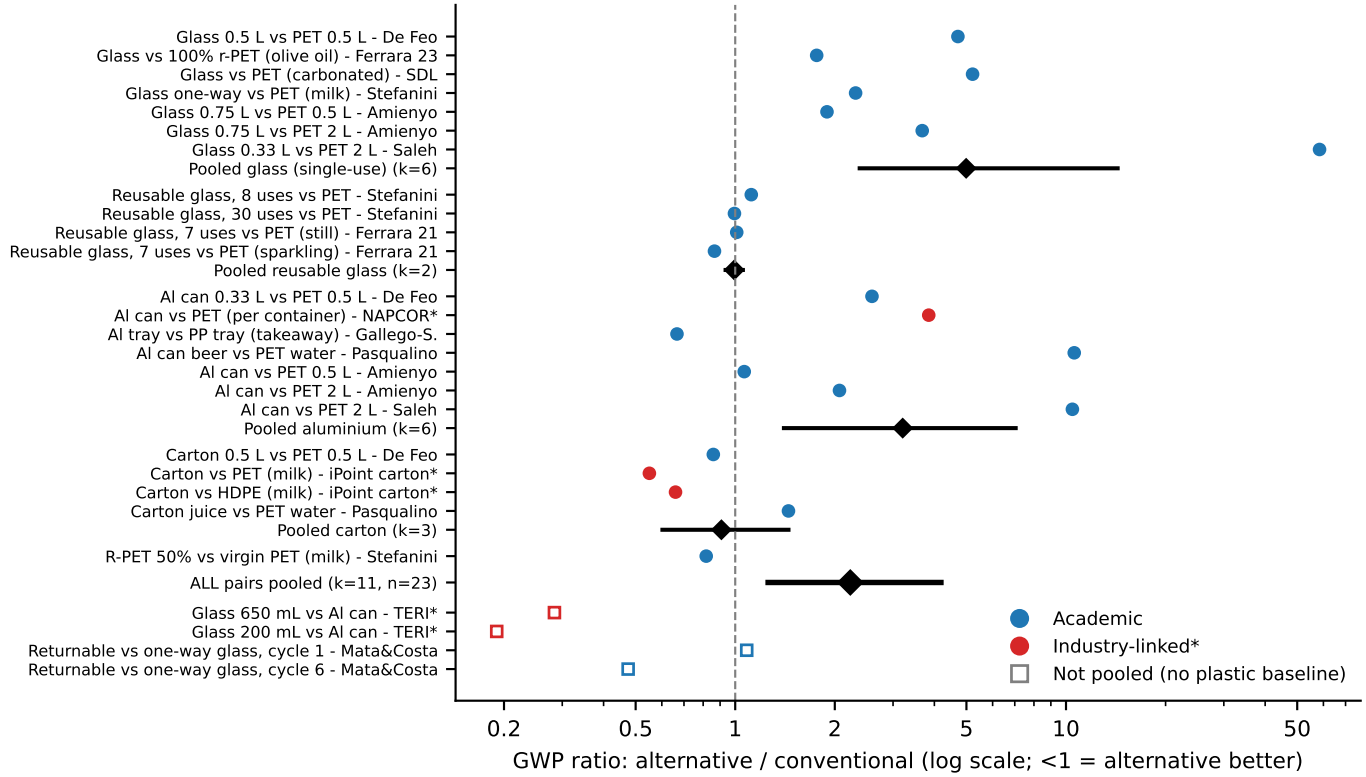


Figure 1: Published GWP ratios (alternative ÷ conventional) and pooled estimates. Blue = academic; red (*) = study linked to a producer or trade body of the winning material (the carton study's funder is not stated in its source); open squares = not pooled; black diamonds = pooled estimates with study-level bootstrap intervals.

3. Results

3.1 As published, alternatives are not generally better (Fig. 1). Across the 23 plastic-baseline pairs the alternative's GWP was 2.2× the conventional one (bootstrap 95% interval 1.26–4.19; t-interval 1.08–4.58; τ = 1.07 in ln units), and the alternative was better in only 30% of pairs. Single-use glass was never better than PET (7 pairs; 5.0×, 2.4–14). Aluminium was worse in 6 of 7 pairs (3.2×, 1.4–7.0). Reusable glass matched PET (0.99×, 0.94–1.06; 7–30

uses). Cartons were better in 3 of 4 pairs (0.91×, 0.60–1.45). In all three industry-linked studies the linked material came out best; academic studies pooled at 2.4× (1.3–5.0).

Robustness: the size of the pooled ratio is not stable. Dropping the extreme Palestinian values gives 1.75×; also dropping the different-beverage pairs gives 1.6× (bootstrap 1.0–2.5); dropping all second-hand values gives 1.56× (bootstrap 0.95–2.6; alternative better in 47% of pairs). The direction (alternatives not better on average) holds, but its size depends on the weakest data.

3.2 Under Indian collection rates the aluminium advantage is marginal (Table 1, Fig. 2). With European rates the model puts aluminium at 1.7× PET and glass at 5.3×, the same direction as the Italian result (2.6× and 4.7×; De Feo et al.). With TERI's Indian rates aluminium's penalty shrinks to 1.15× and it beats PET in 12% of parameter draws; if PET is collected at 90% this falls to 2%. Aluminium reaches parity only above $c^* \approx 89\%$ collection (90% interval 83–96%) against the reported 85% (Fig. 2b). Even in the aluminium-favourable corner (lowest can-making and highest PET-conversion overheads) c^* is 81% (76–86%). Single-use glass stays about 5× PET even at 100% collection (Fig. 2a). The strongest driver of the ratio was the assumed can-making overhead (Spearman $\rho = 0.48$), then PET conversion overhead, aluminium recycling yield and PET resin factor ($|\rho| \approx 0.38$).

Table 1: Model results per litre filled (Monte Carlo, $N = 20{,}000$). "Share of draws" is the fraction of parameter draws, not a probability.

Collection-rate scenario	Al can / PET, median (90% interval)	Share of draws Al < PET	Glass / PET
Europe-default (Al 71%, PET 48%, glass 80%)	1.73 (1.45–2.09)	0%	5.3 (4.2–6.9)
India, TERI rates (Al 85%, PET 70%, glass 60%)	1.15 (0.94–1.43)	12%	5.7 (4.6–7.3)
India, PET collected at 90%	1.29 (1.05–1.62)	2%	6.4 (5.1–8.3)
India, TERI rates, Al-favourable overheads	0.88 (0.75–1.05)	89%	n/a

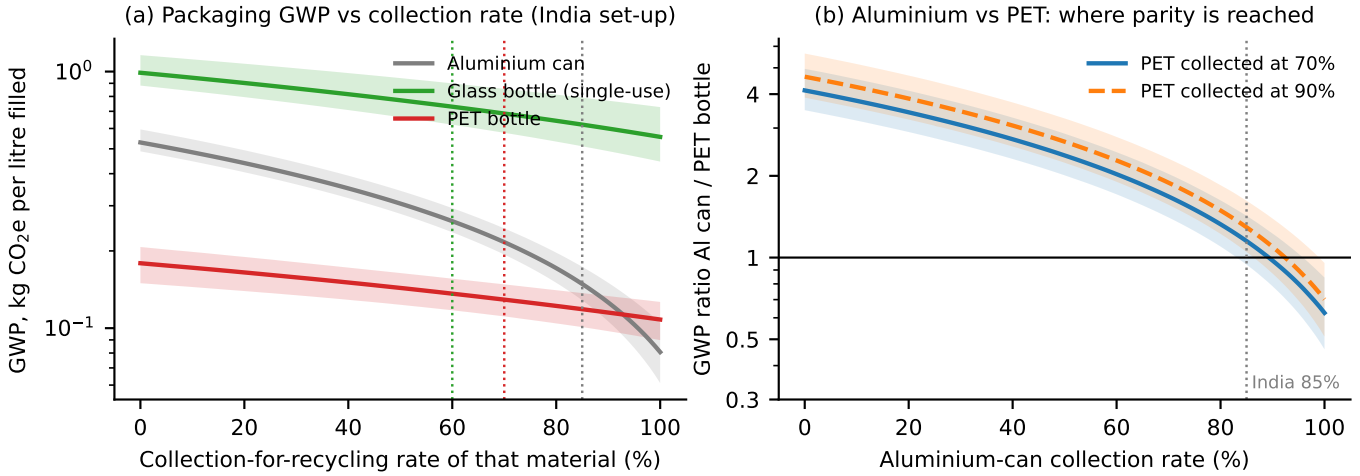


Figure 2: (a) GWP per litre vs. collection rate of each material (India set-up; bands = 5–95% of draws; dotted lines = Indian rates). (b) Aluminium-can ÷ PET ratio vs. aluminium collection rate; parity (ratio = 1) is reached near 89%.

4. Discussion

Carbon advantages of replacement packaging are format-specific and conditional, not general. Cartons are roughly at parity with PET; single-use glass and aluminium are usually worse; reusable glass only matches PET after about 7–30 uses; aluminium wins only when nearly all cans are collected, which is where India's reported 85% sits, so its advantage is fragile rather than wrong. The industry pattern (3 of 3, one unverified) argues for independent comparisons. For glass, reuse rather than recycling is the lever (TERI found five refills cut GWP by 64–73%; Dolci et al. report that single-use PET beats refillable glass below roughly 10 reuses, matching the parity found here); for aluminium, collection investment moves the result more than lightweighting.

Limitations. Only 27 pairs from 13 studies were extractable, fewer than the 25–40 studies (about 60 pairs) targeted, because many studies sit behind paywalls; ten values are second-hand and the Palestinian values look extreme (possible unit issue), so results are shown with and without them. Two Pasqualino pairs compare different beverages; two pairs are per container. Studies were stratified by material and funder, not by functional unit, EoL or allocation method, and the pooled intervals are descriptive. Only GWP and rigid beverage/food containers are covered, not sachets or multilayer flexibles. The Stage 2 model is a screening tool, not an LCA: it omits secondary packaging, uses three unsourced overheads, and overstates glass against aluminium in India (model 4.9× vs. TERI 3.5×) and does not reproduce TERI's ranking of aluminium ahead of PET, so c^* is indicative.

5. Conclusion

The claim "sustainable packaging reduces carbon" does not hold in general: in the published pairs, replacements for plastic were better in only about one pair in three, and the size of the average penalty depends on the weakest data. Under Indian collection rates the one clear contender, the aluminium can, sits close to parity with PET and needs about 89% collection to match it, so it is a fragile advantage rather than a sure one. Single-use glass does not compete at any collection rate.

Files: `UNWRAP_data_and_results.xlsx` (S7 references, S8 added checks, S9 live model), `analysis.py`, `paired_comparisons.csv`.